MILITARY SPECIFICATION

RADIO SET AN/ARC-124(*)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall provide radio frequency reception and transmission facilities for data communications systems such as the Digital Data Communication Set AN/ASW-21A and AN/ASW-27(*). Operationally, the equipment shall provide two-way communication of digital data information on a time division basis by means of frequency shift keyed type of modulation.

1.2 Classification. The equipment covered by this specification shall consist of the following items:

Item	*Type Designation*	*Appl. Para.*
Receiver-Transmitter, Radio	RT-803(*)/ ARC-124	3.4.1.
Control, Radio Set	C-6867(*)/ ARC-124	3.4.2
Mounting	MT-3598(*)/ ARC-124	3.4.3

1.3 Associated Equipment. The equipment shall operate with the following associated equipment which shall not be supplied as part of this specification:

Antenna	AT-141A/AR or equivalent with RG-8/U cable.
Digital Data Communication Set	AN/ASW-21A or AN/ASW-27(*)
Integrated Electronic Central Control	C-4850/ASQ-85

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-172 — Cases; Bases, Mounting; and Mounts, Vibration (For Use With Electronic Equipment in Aircraft)

MIL-E-4682 — Electron Tubes and Transistors, Choice and Application of

MIL-W-5088 — Wiring; Aircraft, Installation of

MIL-E-5400 — Electronic Equipment, Aircraft, General Specification for

MIL-T-5422 — Testing, Environmental, Aircraft Electronic Equipment

MIL-I-6181 — Interference, Control Requirements, Aircraft Equipment

MIL-C-6781 — Control Panel: Aircraft Equipment, Rack or Console Mounted

FSC 5821

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MIL-P-7788 — Plate, Plastic, Lighting

MIL-M-7793 — Meter, Time Totalizing

MIL-E-17555 — Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

MIL-T-18303 — Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for

MIL-N-18307 — Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-E-19600 — Electronic Modules, Aircraft, General Requirements for

MIL-R-22256 — Reliability Requirements for Design of Electronic Equipment or Systems

MIL-R-23094 — Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

MIL-T-23103 — Thermal Performance Evaluation, Airborne Electronic Equipment, General Requirements for

MIL-D-23456 — Digital Data Communications Set AN/ASW-21

MIL-S-23603 — System Readiness/Maintainability; Avionics System Design, General Specification for

MIL-C-26482 — Connector, Electric, Circular, Miniature, Quick Disconnect

MIL-D-81170 — Digital Data Communications Set AN/ASW-27(*)

STANDARDS

MILITARY

MIL-STD-704 — Electric Power, Aircraft, Characteristics and Utilization of

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

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(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL–E–5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL–E–5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.3)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 *Nonstandard Parts and Materials Approval.* Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL–E–5400.

3.1.2 *Electron Devices.* Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL–E–4682.

3.1.3 *Microelectronic Modular Assemblies.* When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency. Microelectronic Modular Assemblies (Microelectronic Circuits) (MICS) shall be contained within Maintenance Modules (for example, on printed circuit cards). The choice of Microelectronic Modular Assemblies shall be based on at least two sources of supply, which can be used interchangeably.

3.1.4 *Modules, Maintenance.* The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL–S–23603). Maintenance modules shall normally be considered repairable.

3.1.5 *Microelectronic Interchangeability.* Microelectronic circuits (MICS) shall be interchangeable at the MIC level, or at the Maintenance module level when approved by the procuring activity. (See 6.9)

3.1.6 *Switches.* Rotary switches mechanically positioned by other than manual means shall be mounted by standoffs.

3.2 Design and Construction. The equipment shall conform with all the applicable requirements of Specification MIL–E–5400 for design, construction, and workmanship, unless otherwise specified herein.

3.2.1 *Cooling.* Two copies of the cooling design data required by Specification MIL–E–5400 and the results of the thermal evaluation shall be submitted to the Bureau of Naval Weapons, Code RAAV–331, for review. The thermal test shall be in accordance with Specification MIL–T–23103. The data from this test may be used to supply the information called for in the reliability thermal survey.

3.2.2 *Total Weight.* The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 15 pounds.

3.2.3 *Operating Life.*

3.2.3.1 *Operational Stability.* The equipment shall operate with optimum performance for 1000 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 *Total Operating Life.* The equipment shall have a minimum total operating

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life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 *Reliability in Mean Time Between Failures (MTBF).* The equipment shall have 1000 hours of mean (operating) time between failures at a message rate of 10 messages per second, when tested and accepted as outlined under the requirements of 4.4.3. The equipment shall be planned, designed, analyzed, and the reliability reported as outlined in Specification MIL–R–22256.

3.2.4 *Time Totalizing Meter.* The receiver-transmitter shall contain a time totalizing meter in accordance with Specification MIL–M–7793.

3.2.5 *Cabling and Connection.*

3.2.5.1 *Cables and Connectors.* The equipment shall provide for the use of cables and connectors in accordance with Specification MIL–E–5400.

3.2.5.2 *Interconnection Cabling.* The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL–W–5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.6 *Nomenclature and Nameplates.* Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL–N–18307.

3.2.7 *Interference Control.* The generation of radio interference by the radio set and the vulnerability of the radio set to radio interference shall be controlled within the limits of Specification MIL–I–6181.

3.2.8 *Interchangeability of Reordered Equipment.* For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.9 *Maintenance Provisions and Field Testing.* Provisions for maintenance shall be as specified in Specification MIL–E–5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.10 *Overload Protection.* Overload protection shall be provided in accordance with Specification MIL–E–5400.

3.2.11 *Undervoltage Protection.* The equipment shall not be damaged by voltages below the minimum specified in MIL–STD–704, and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 *Control Panels.* All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL–C–6781 for Type I Control Panels. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.13 *Standard Conditions.* The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

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Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC

3.2.14 *Service Conditions.* The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL–E–5400 for Class 2 equipment without cooling air.

3.2.14.1 *Vibration.* The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL–E–5400. The equipment shall be designed for hard mounting without vibration isolators.

3.2.14.2 *Acoustic Noise Test.* No malfunction of the equipment shall occur when subjected to a jet engine exhaust noise of 120 db relative to 0.0002 dyne per square centimeter at frequencies between 20 and 10,000 cps.

3.2.15 *Pressurization.* Items of equipment to be sealed shall comply with Specification MIL–E–5400 except that a safety valve is not required, mechanical pressurization pumps shall not be provided in the equipment and the cases shall be designed to be pumped with dry air to a differential of 2 to 4 pounds per square inch at sea level pressures and room ambient temperatures. Loss of pressure shall not exceed 5 pounds per square inch for a 24 hour period at 100,000 feet altitude and +55°C ambient temperature.

3.2.15.1 *Air Pressure Indicator.* A non-air-release type pressure indicator shall be mounted in the front of the cover of the receiver-transmitter and shall give a visual indication when the internal air pressure is adequate for proper operation of the equipment.

3.2.16 *Primary Input Power Requirements.* The equipment shall meet all applicable requirements of MIL–STD–704 and shall give specified performance from the following power sources with characteristics as defined in MIL–STD–704 having limits as specified therein. The power required shall not exceed the specified amounts. Application of primary power to the equipment will be controlled by the associated digital communications system.

(1) AC Power (Three Phase) 115/200V, Category B, 175VA

3.2.17 *Warm-Up-Time.* The warm-up time for the equipment shall not exceed 1 minute under standard conditions and 3 minutes at extreme service conditions. After warm-up the equipment shall be capable of meeting the requirements specified herein.

3.3 Performance.

3.3.1 *General.* Unless otherwise specified, values set forth to establish requirements for satisfactory performance apply under both standard and service conditions. Reduced performance and service requirements shall apply only where specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated continuously to receive and transmit 10 messages per second in accordance with the message timing standards specified in Specification MIL–D–23456 and Specification MIL–D–81170.

3.3.2 *Frequency Channels.* By means of control box selection, it shall be possible to select anyone of 3500 frequencies in the range of 225.00 to 399.95 mc in 0.05 mc increments. These main channel frequencies shall correspond with the arithmetic series 225.00, 225.05, 225.10,———399.95mc.

3.3.3 *Terminations.*

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3.3.3.1 *Receiver Input.* The impedance at the receiver input terminals shall be 52 ohms resistive.

3.3.3.2 *Transmitter Output.* The impedance at the transmitter output terminals shall be 52 ohms resistive.

3.3.3.3 *Receiver Output.* The data output of the receiver shall be terminated in a 2000-ohms non-inductive load.

3.3.3.4 *Transmitter Data Input.* The data input circuit of the transmitter shall present a resistive load of nominally 2500 ohms to the signal source.

3.3.3.5 *Turn-around and Carrier-on Pulse Terminations.*

(a) *Turn-around Input.* The transmitter turn-around input circuit shall present a load of not less than 100,000 ohms to the keying source.

(b) *Carrier-on Input.* The carrier-on input circuit of the transmitter shall present a load of not less than 100,000 ohms to the keying source.

3.3.4 *Receiver Sensitivity.* Sensitivity of the receiver under standard conditions shall be such as to produce 20 db of quieting for a 3 microvolt (open circuit microvolts) input cw signal. Quieting shall be measured at the data output terminals through a low pass filter providing attenuation of 3 db at 10 kc. A signal level of not more than 5 microvolts (open circuit) shall produce 20 db of quieting under service conditions.

3.3.5 *Selectivity.* The overall selectivity of the receiver shall fall within the limits set forth below:

Attenuation Below Resonant Response	*Total Bandwith*
6 db	86 kc min.
60 db	180 kc max.

Peak to valley ratio shall be not more than 1 db over the 90 kc bandwith.

3.3.6 *Regeneration.* The interstage and overall regeneration of the receiver shall be so minimized that no evidence of instability, unnecessary oscillation, or excessive regeneration shall occur at maximum gain under standard or service conditions.

3.3.7 *Receiver Overall Frequency Responses.* The response shall not vary more than 5% from the maximum data signal over a frequency range from dc to 7,000 cps with sine wave frequency modulation and ± 20 kc deviation. Test equipment frequency linearity error shall not be included in the 5% limit.

3.3.8 *Receiver Output Linearity.* The data output voltage shall be linear within ± 5% of the maximum peak-to-peak value data output voltage for signal deviation from zero up to ± 20 kc.

3.3.9 *Receiver Amplitude Variation.* With a constant data input signal at ± 20 kc deviation at a 1000 cps rate, the output shall not vary more than 10% for RF signal level variation from limiter saturation to 20 db above this level. (Limiter shall be saturated with signal input to the receiver of — 80 dbm.)

3.3.10 *Receiven Output Capability.* The receiver shall be capable of delivering an output of at least 10 volts peak-to-peak into the data output load with a signal having an RF level sufficient for limiter saturation and frequency modulated sinusoidally at ± 20 kc deviation with a 1000 cps rate.

3.3.11 *Receiver Internal Noise Level.* The internal noise measured at the data output terminals through the low pass filter described in 3.3.4 shall not exceed 5% of the peak-to-peak data signal when an unmodulated RF signal of —85 dbm is applied to the receiver.

3.3.12 *RF Power Output.* The transmitter shall be capable of delivering the follow-

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ing values of R F carrier energy within the tuning range:

Standard Conditions	20 watts minimum
Service Conditions	15 watts minimum
Reduced Performance Test at 102 vac and 22 vdc	10 watts minimum

3.3.13 *Modulation Characteristics:* The transmitter shall meet the following performance requirements for modulation.

3.3.13.1 *Type of Modulation.* The transmitter shall employ frequency shift modulation.

3.3.13.2 *Capability.* The modulation capability shall be such that +5.0 volts applied to the data input terminals shall increase the carrier frequency 20 kc (±5%) and —5.0 volts applied to the data input terminals shall decrease the carrier frequency 20 kc (±5%).

3.3.13.3 *Transmitter Noise Level.* Without modulation, the frequency deviation developed by internal noise shall not exceed 5% of the maximum peak-to-peak deviation of the data signal.

3.3.13.4 *Transmitter Frequency Linearity.* The FSK modulated RF frequency shall be linear within ±5% of the maximum peak-to-peak frequency deviation over a range of data input voltage levels between 0 volts and ± 5.0 volts.

3.3.13.5 *Transmitter Frequency Response.* The peak-to-peak frequency deviation shall not vary more than 5% with a sine wave modulated signal applied to the data input terminals of constant level and over the frequency range from 0 to 7000 cps. Test equipment frequency linearity error shall not be included in the 5% limit.

3.3.14 *Frequency Stability.* The frequency stability of the complete equipment after warm-up shall be such as to maintain the equipment frequency within the limits shown below provided the external ambient temperature does not exceed 71°C.

Transmitter Carrier Frequency Tolerances	± 5 kc
Receiver Resonant Frequency Tolerance	± 5 kc

3.3.15 *Channel Changing Cycle.* The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than 1 second. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels.

3.3.16 *Accuracy of Automatic Tuning.* When the same frequency channel is selected several times in succession, the transmitter RF power output and the receiver output power (with an input signal of 3 microvolts or greater with a 20 kc deviation at 1000 cps) shall not vary more than 0.5 db from the power output obtained initially.

3.3.17 *Cable Length.* Under the standard or service conditions, the equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from #22 AWG copper wire, or when using cables up to 110 feet in length fabricated from #16 AWG wire.

3.3.18 *Receive to Transmit Switching Requirements.* The equipment shall meet the following requirements for switching from receive to transmit operation.

3.3.18.1 *Receive to Turn-around Operation.* The equipment shall remain in receive condition when —13 to —30 volts is applied to the Turn-around input circuit. Application of a Turn-around pulse of 0 to —2 volt level shall cause the equipment to be

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switched to the prepared-to-transmit condition.

3.3.18.2 *Turn-around Keying Time.* Turn-around operation, from receive to prepared to transmit condition, shall not exceed 28 milliseconds after application of the Turn-around pulse to the input circuit. Turn-around operation from transmit to maximum sensitivity receive condition shall not exceed 28 milliseconds.

3.3.18.3 *Leakage Power.* Leakage RF power from the transmitter shall not exceed —85 dbm at the operating frequency when in Turn-around condition.

3.3.18.4 *Carrier-on Operation.* From Turn-around condition, the transmitter shall be capable of being keyed upon application of a keying voltage having the following characteristics. The magnitude of the keying pulse shall be not less than 13 volts peak-to-peak centered at 0 volts maximum peak deviation of the voltage shall be within +11 volts and —30 volts. The carrier shall be on during the positive portion of the keying voltage. An external short circuit between the keying voltage input terminal of the external connector and ground shall not cause the transmitter to be continuously keyed.

3.3.18.5 *Carrier on Keying Time.* Maximum specified RF power shall be applied to the output load within 160 microseconds after application of each Carrier-on pulse. The RF power in the output load shall decrease to not more than —85 dbm within 80 microseconds after cessation of the Carrier-on pulse.

3.3.19 *External Wiring.* The receiver-transmitter shall be designed to use the same external wiring as Receiver-Transmitter RT–649/ARC–88 except that an additional wire may be required as indicated in 3.4.1.3.

3.4 Detail Requirements.

3.4.1 *Receiver-Transmitter RT–803/ARC–124.* The receiver-transmitter shall meet the following requirements.

3.4.1.1 *Function.* The receiver portion of the receiver-transmitter shall receive frequency shift keyed energy from the antenna, amplify and convert it to digital type information to be applied to data multiplex equipment. The transmitter portion shall generate RF power, frequency shift keyed by digital information supplied in time division sequence from data multiplex equipment, and transmit it to the antenna.

3.4.1.2 *Form Factor and Mounting.* The receiver-transmitter shall be contained in a single sealed case which meets the applicable requirements of Specification MIL–C–172 and capable of being mounted in an aircraft by means of Mounting MT–3598/ARC–124. It shall be possible to dismount the unit and remove it from its case without the use of tools and without requiring removal of any cables other than the coaxial and power connectors. The maximum volume shall not exceed 300 cubic inches.

3.4.1.3 *Contents.* The receiver-transmitter shall contain the complete transmitter, FSK modulator, receiver, frequency generator, frequency selection system, automatic tuning devices, means for measuring incident and reflected power, and all other parts to make the equipment function except the radio set control. It shall be possible to replace any plug-in unit with another without readjustment or use of hand tools. A wire shall be connected between the external connector and the appropriate module to permit application of an external bias to desensitize the receiver for test purposes.

3.4.1.4 *Weight.* The weight of the receiver-transmitter shall not exceed 12.75 pounds.

3.4.1.5 *Electrical Connections.*

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3.4.1.5.1 *External.* The following electrical connections shall be furnished on the base of the receiver-transmitter:

(1) External connectors, as required, shall conform with Specification MIL–C–26482.

(2) Test points as required on the front panel to facilitate maintenance procedures.

3.4.1.6 *Internal Controls.* No internal controls shall be required for line maintenance. Semi-fixed or adjustable controls necessary to meet the performance requirements of this specification shall be an integral part of each card or maintenance module and shall be preset at the factory or shop levels.

3.4.1.7 *External Controls.* The following controls shall be located on the front of the receiver-transmitter.

(1) Receiver zero reference control.

(2) Transmitter data input level control.

(3) Momentary contact switch for Turn-around test.

(4) Momentary contact switch for Carrier-on test.

(5) Momentary contact switch to be depressed for measuring reflected power.

3.4.2 *Control, Radio Set C–6867/ARC–124.* This control shall conform to the applicable requirements of Specification MIL–C–6781 and shall be suitable for installation in a console panel.

3.4.2.1 *Function.* This unit shall provide all controls for frequency selection of the receiver-transmitter. It shall be possible to manually select any of the 3500 available frequencies and preset up to 20 channels on any of these available frequencies. Lighting panels shall conform with Specification MIL–P–7788 for class 1–R, Type IV panels.

3.4.2.2 *Form Factor and Mounting.* The over-all dimensions shall not exceed 5¾ inches in width, 4⅞ inches in length and 5¾ inches in depth.

3.4.2.3 *Weight.* The weight of the control, including the mating connector, shall not exceed 2.0 pounds.

3.4.2.4 *Contents.* The control shall contain all circuits necessary to control manual frequency or preset channel selection. Solid state tuning shall be used. Rotary motion shall not be required to operate any of the controls.

3.4.2.5 *Electrical Connections.* All electrical connections shall be made through a connector conforming to Specification MIL–C–26482.

3.4.2.6 *Controls, Panel-Mounted.* The following controls shall be located and suitably marked on the front panel.

(1) A 20-position switch-type control CHAN for selecting any one of 20 predetermined main frequencies.

(2) Controls to operate a digital display labeled MEGACYCLES for manually selecting any one of 3500 channels. The frequency readout shall indicate the frequency on which the equipment is operating when in the MAN position.

(3) A 2-position switch to allow rapid selection of two operating frequencies as follows: Position 1—PRESET, for operating on any one of the predetermined frequencies selected by (1) above; position 2—MAN, for

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operation on any of the manually selected frequencies determined by (2) above.

(4) Means for readily setting up new predetermined frequencies on the 20-position switch type control while in flight by selecting the frequency manually and pressing a momentary contact CHAN SET control.

3.4.3 *Mounting MT–3598/ARC–124.* The mounting shall meet the following requirements:

3.4.3.1 *Function.* The mounting shall provide a means of securing the receiver-transmitter to a plane surface within the aircraft or vehicle.

3.4.3.2 *Form Factor and Mounting.* The mounting shall be a sheet metal template with hold-down hardware. The overall dimensions of the mounting shall be specified by the manufacturer.

3.4.3.3 *Weight.* The weight of the mounting shall not exceed 0.25 pounds.

3.5 Design Data. No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order. (See 6.6)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Classification of Tests.* Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests.

(2) Initial Production Tests.

(3) Acceptance Tests.

(4) Life Tests.

4.2 Preproduction Tests. Preproduction tests shall be made on an equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 *Preproduction Test Data.* The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme temperature operating condition.

4.2.2 *Scope of Tests.* Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Speci-

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fication MIL–T–5422 and interference tests in accordance with Specification MIL–I–6181.

4.2.3 *Preproduction Approval.* Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 *Production Equipments.* Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary correction as approved by the procuring activity.

4.3 Initial Production Tests. One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 *Scope of Tests.* This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 *Accessory Material.* In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 *Initial Production Sample Approval.* Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 *Reconditioning of Initial Production Test Sample.* On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damage items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable.

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Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests.

(2) Sampling Tests.

(3) Reliability Assurance Tests.

(4) Special Tests.

4.4.1 *Individual Tests.* Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product.

(2) Operational Test.

(3) Manufacturing Run In Test.

4.4.1.1 *Examination of Product.* Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 *Operational Test.* Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 *Manufacturing Run In Test.* Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 *Sampling Tests.* Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with

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the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	*Quantity to be Selected For Sampling Test*
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 *Scope of Tests.* As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 *Reliability Assurance Tests.* Reliability Assurance Tests shall be as required by Specification MIL–R–23094, Procedure I, Equipments selected for reliability assurance tests shall first have passed the individual tests and shall be selected at random by a representative of the procuring activity.

4.4.3.1 *Test Level.* The Test Level shall be Test Level III (from Table I of Specification MIL–R–23094), except that the upper temperature-limit shall be +71°C.

4.4.3.2 *Accept-Reject Criteria.* Figure 11 of Specification MIL–R–23094 and an MTBF of 1000 hours shall be used for the sample plan.

4.4.3.3 *Length of Heat Portion of Cycle.* The length of the heating cycle, during which time the equipment shall be operated, shall be recommended by the contractor, and be subject to approval of the procuring activity.

4.4.3.4 *Performance Requirements During Test.* The performance characteristics to be measured and the minimum performance requirements to be met during the tests shall be as follows:

Characteristic	*Performance requirement*
R. F. Power Output	3.3.12 (service conditions)
R. F. Output Modulation	3.3.13.2
Receiver Sensitivity	3.3.4 (service conditions)
Receiver Data Output	3.3.10

4.4.3.5 *Failure Criteria.* In addition to the requirements of Specification MIL–R–23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated,

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each shall be counted a failure, unless one caused the other parts to fail.

4.4.4 *Special Tests.* Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 *Special Test Schedule.* Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 *Scope of Tests.* Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 *Equipment Failure.* Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 1000-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 1000 hours each of test time. The life test on the radio set control shall in addition consist of operating each control knob for a total of 5000 cycles throughout its complete range.

Quantity of Equipments Offered for Acceptance	*Quantity to be Selected for Life Test*
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 *Test Conditions.* The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0–5000 ft.)

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Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)

4.5.2 *Test Periods.* The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 *Performance Check.* At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 *Test Data.* The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number.

(2) The circuit reference symbol number.

(3) The part function.

(4) Name of the manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.5.4.1 *Failure Report.* In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 *Reconditioning of Life Test Samples.* An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all wear items. The wear items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures. The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL–T–18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment. Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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4.8 Presubmission Testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest. Equipment which has been rejected may be reworked or have parts replaced to correct defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL–E–17555.

6. NOTES

6.1 Test Values. Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5V DC and 115 ± 1.0V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.2 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL–E–5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) *Contract.* The contract shall have precedence over any specification.

(2) *This Specification.* This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) *Referenced Specifications.* Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.5 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.5).

6.7 Asterisk. In specification revisions and superseding amendments an asterisk (*) preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Criteria for Multiple Source. The criteria of this paragraph shall be used for determining the extent to which the requirement for multiple independent source availability of microcircuits is satisfied. For interchangeability at the MIC level, multiple source availability exists if the devices from the different sources may be used interchangeably, device by device, in the final equipment without any need for mechanical or electrical changes. For interchangeability at the card level, multiple source availability exists if a card composed of MICS from one source is mechanically and electrically interchangeable with another card composed of MICS from another source. Numerous aspects need be considered to arrive at the independence of sources; these involve production independence, corporate (or firm) and financial independence, and geographical separation. The production independence of one source of another is determined by the extent to which the one source is independent of the other for manufacture; and by the lack of common original suppliers of raw or processed materials. Design, process information, technology, and diffusion masks may be derived from a common source. However, in such a case, each device source shall have a demonstrated capability of origination these items. The corporate (or firm) and financial independence of one source of another is determined by the absence of common directors, common officers, or holders of large common financial interests. Geographical separation is simply the distance between the locations where principal manufacture takes place by two sources.

6.9 Microelectronic Interchangeability Demonstration. Interchangeability of microelectronic elements shall be demonstrated as follows:

(1) For the interchangeability at the MIC level, it shall be demonstrated for each type of functional MIC from one source, as used in the system, that is mechanically and electrically interchangeable with that from another source (see paragraph 6.8).

(2) For the interchangeability at the Maintenance module level, it shall be demonstrated that modules typical of those to be used in the system composed of MICS from one source are mechanically and electrically interchangeable with modules composed of MICS from another source.

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(3) For the completed equipment, it shall be demonstrated that modules composed of MICS from one source are interchangeable with corresponding modules composed of MICS from another source. When interchange has been effected, no degradation shall take place in system and subsystem performance.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81299(WP) RADIO SET AN/ARC-124(*)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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